



# DAYANANDA SAGAR UNIVERSITY

Devarakaggalahalli, Harohalli, Kanakapura Road,  
Bengaluru South Dt. – 562 112, Karnataka, India



**SCHOOL OF  
ENGINEERING**

An Internship Report on

**Backend Developer**

*Submitted in Partial fulfillment for award of degree in*

**Bachelor of Technology  
in  
COMPUTER SCIENCE AND ENGINEERING**

*Submitted by*

**RAVI THARUN**

**ENG22CS0407**

Internship carried out at

**Rebert Technologies Private. Limited**

**INTERNAL GUIDE**

**Dr. Sivananda Lahari Reddy**  
Associate Professor  
Dept. of CSE, School of  
Engineering  
Dayananda Sagar University

**EXTERNAL GUIDE**

**Ms. Sara Shaikh, Chief  
Technology Officer,**  
Rebert Technologies Private. Limited, Prasad No  
7, Sadicha Apartment, Yerawada, Pune,  
Maharashtra - 411006

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING,  
SCHOOL OF ENGINEERING  
DAYANANDA SAGAR UNIVERSITY,  
(2025- 2026)**



# DAYANANDA SAGAR UNIVERSITY



SCHOOL OF  
ENGINEERING

## Department of Computer Science & Engineering

Devarakaggalahalli, Harohalli, Kanakapura Road,  
Bengaluru South Dt. – 562112

### CERTIFICATE

This is to certify that the Internship report, entitled “**Backend Developer**” by **Ravi Tharun (ENG22CS0407)**, a Bonafide student of Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment of the requirement for the during the academic year 2025-26. It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the report. The Internship report has been approved as it satisfies the academic requirements in respect of Internship prescribed for the VIII Semester.

**External Guide**  
Ms. Sara Shaikh  
Chief Technology Officer,  
Rebert Technologies  
Private. Limited, No 7,  
Sadicha APartment, Pune  
Maharastra -411006

**Internal Guide**  
Dr. Sivananda Lahari Reddy  
Associate Professor  
Dept. of CSE,  
School of Engineering  
Dayananda Sagar University

**Dr Girisha G S**  
Chairman CSE  
School of Engineering  
Dayananda Sagar University

Date:

23/04/26

Date:

Date:

Name of The Examiner

Signature of Examiner

1.

2.



# DAYANANDA SAGAR UNIVERSITY

**Department of Computer Science & Engineering**  
Devarakaggalahalli, Harohalli, Kanakapura Road,  
Bengaluru South Dt. – 562112



**SCHOOL OF  
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## DECLARATION

I, **RAVI THARUN**, hereby declare that the Internship titled “**Backend Developer**” embodied in this report has been carried out by me during VIII Semester B. Tech in **Computer Science and Engineering**, at School of Engineering, **Dayananda Sagar University**, submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Computer Science and Engineering** during the academic year **2025-2026**. The work embodied in this report is original & it has not been submitted in part or full for any other degree in any University.

**RAVI THARUN**  
**ENG22CS0407**

---

**Place: Bangalore**

**Date:**

## ACKNOWLEDGEMENT

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*I would like to thank our External guide **Ms. Sara Shaikh, Chief Technologies Private Limited, Pune, Maharashtra**, for sparing his/her valuable time to extend help in every step of our Internship work, which paved the way for smooth progress and fruitful culmination of Internship.*

*I would like to thank our Internal guide **Dr.Sivananda Lahari Reddy, Assistant Professor, Dept. of Computer Science and Engineering, Dayananda Sagar University**, for sparing his/her valuable time to extend help in every step of our Internship work, which paved the way for smooth progress and fruitful culmination of the Internship.*

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**Ravi Tharun**  
**ENG22CS0407**

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# **ABSTRACT**

This report presents the work carried out during the internship in the domain of backend development using the MERN stack. The primary objective of the project was to design and develop a course selling website backend system that supports user management, course handling, and order processing through secure and efficient API.

The backend system is developed using Node.js and Express.js, with MongoDB used as the database for storing and managing application data. The system focuses on implementing RESTful APIs, handling authentication and authorization, and ensuring smooth communication between client and server.

A key contribution during the internship was the development of backend functionalities such as user registration and login, course creation and management, and purchase/order handling. The APIs were designed to ensure secure data flow and efficient processing of requests.

The developed backend system demonstrates how scalable and efficient server-side applications are built for real-world e-learning platforms. Overall, the internship provided strong practical exposure to backend development, API design, and database integration.

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# CHAPTER 1 ABOUT THE ORGANIZATION

## 1.1. INTRODUCTION:

**Rebert Technologies Private Limited** is a technology-driven IT solutions company that delivers innovative digital products and services across web development, mobile applications, and enterprise software solutions. The company focuses on building scalable, reliable, and high-performance solutions for clients across India.

Rebert Technologies operates with a strong emphasis on modern technologies such as Artificial Intelligence, Machine Learning, Cloud Computing, DevOps, Blockchain, IoT, Cyber Security, and Data Science. By combining technical expertise with creative innovation, the company ensures end-to-end digital transformation for businesses and startups.

The organization follows a client-centric development approach, ensuring timely execution, quality delivery, and continuous support throughout the project lifecycle. It also plays a key role in industrial training, internships, and technical skill development for students and professionals.

With a strong team of designers, developers, and technology enthusiasts, Rebert Technologies continues to expand its presence in the IT industry by delivering impactful solutions in areas like product development, digital transformation, HR outsourcing, and IT Services.

## **1.2. VISION**

Rebert Technologies envisions becoming a leading global IT solutions provider by delivering innovative, scalable, and technology-driven services. The organization aims to empower businesses through advanced digital solutions, creating long-term value and enhancing customer experience across industries.

## **1.3. MISSION**

Rebert Technologies' mission is to provide innovative and reliable IT solutions that help businesses grow through technology. The company is committed to:

- Delivering high-quality and customized digital solutions
- Improving efficiency, performance, and scalability of business systems
- Continuously adopting and integrating modern technologies like AI, ML, Cloud, and DevOps
- Providing end-to-end support for web, mobile, and enterprise application development By taking full ownership of operational challenges, Rebert Technologies ensures consistent, reliable, efficient outcomes for its clients
-

## 1.4. TEAM

**Rebert Technologies** workforce forms the backbone of the organization, consisting of skilled designers developers, and technology professionals who contribute to innovation and operational excellence. The company promotes a culture of continuous learning, collaboration, and growth.

A key focus of Rebert Technologies is supporting employee development through training, mentorship, and real-time project experience

The organization provides:

- **Skill Development Programs:** Designed to improve technical and professional skills through workshops, training sessions, and industry-relevant learning .
- **On -the-Job Training:** Focuses on practical learning by working on real-time projects under the guidance of experienced professionals.
- **Mentorship & Coaching:** Senior team members support junior developers by sharing knowledge, improving skills, and helping them achieve career growth .

## 1.5. SERVICES

Rebert Technologies offers a comprehensive range of IT services that support digital transformation and improve business performance:

**Product Design & Development:** Provides end-to-end development of web applications, mobile applications, and software solutions using modern technologies to deliver scalable and efficient products.

**AI, ML & Emerging Technologies Solutions:** Delivers intelligent solutions using Artificial Intelligence, Machine Learning, IoT, Cloud Computing, DevOps, Blockchain, AR/VR, Data Science, ERP, and CRM systems to enhance business automation and decision-making.

**Complete Website & Mobile App Development:** Builds responsive, user-friendly, and high-performance websites and mobile applications tailored to client requirements.

**Data Management & IT Security Solutions:** Ensures secure handling of data with efficient storage, management, and cybersecurity solutions.

**HR Staffing & Outsourcing Solutions:** Offers skilled IT professionals and workforce solutions to help companies manage technical resources effectively.

## **CHAPTER 2 ABOUT THE DEPARTMENT**

### **2.1. INTRODUCTION**

The Department of Information Technology at Rebert Technologies focuses on designing, developing, and maintaining modern software solutions that meet industry needs. It plays a key role in building scalable web applications, mobile applications, and enterprise-level systems using current technologies.

The department is responsible for handling end-to-end project development, including requirement analysis, system design, development, testing, and deployment. It ensures that all solutions are efficient, secure, and user-friendly.

With a strong team of developers and designers, the department continuously works on improving technical skills, adopting new technologies, and delivering high-quality digital products that support business growth and innovation.

## **2.2. EMPLOYEE EXPERIENCE DEPARTMENT**

The internship provided a valuable and enriching experience within Rebert Technologies as a Remote Backend Developer. Unlike academic learning, which is mainly theoretical, this remote work environment offered practical exposure to real-world backend development and distributed teamwork.

During the internship, hands-on experience was gained in backend development, API creation, and database handling. This helped in understanding how server-side applications are designed, developed, and integrated with frontend systems in real-time projects, even in a remote setup.

The work environment was highly collaborative and supportive despite being remote. Communication with team members was maintained through online platforms, and guidance was provided whenever required. This encouraged continuous learning, improved technical understanding, and helped in solving problems effectively.

In addition to technical learning, the internship also helped in understanding professional work ethics in a remote environment, communication practices, and teamwork, which are essential in modern distributed software development corporate setting.

## 2.3. DAILY MEETINGS

During the internship, regular daily meetings were conducted to track progress, assign tasks, and review completed work. These meetings played an important role in ensuring smooth workflow and effective communication within the team.

Each day, updates were shared on **yesterday's progress**, followed by discussion of **today's tasks and goals**. This helped in clearly understanding what work was completed and what needed to be done next.

In addition to task updates, **code reviews** were also conducted where the work done was checked, and suggestions or improvements were provided. This helped in improving code quality and following proper development standards.

The meetings also allowed team members to clarify doubts, discuss issues, and get guidance from seniors whenever required. This improved learning and helped in completing tasks more efficiently.

Overall, daily meetings helped in tracking progress, improving coordination, and ensuring steady development throughout the internship.

## **2.4. TECH TALKS**

The organization regularly conducts technical discussions known as Tech Talks. These sessions are used to introduce new tools, software, and technologies, along with explaining how to use them in real projects..

During these sessions, I was informed about different development tools, platforms, and software used in the industry. The team also explained how to use these tools effectively in backend development and project workflows.

These Tech Talks helped me understand practical usage of new technologies and improved my technical knowledge. They also made it easier to adapt to new tools and apply them in daily development tasks.

Overall, Tech Talks played an important role in improving my learning experience by keeping me updated with modern tools and industry practices



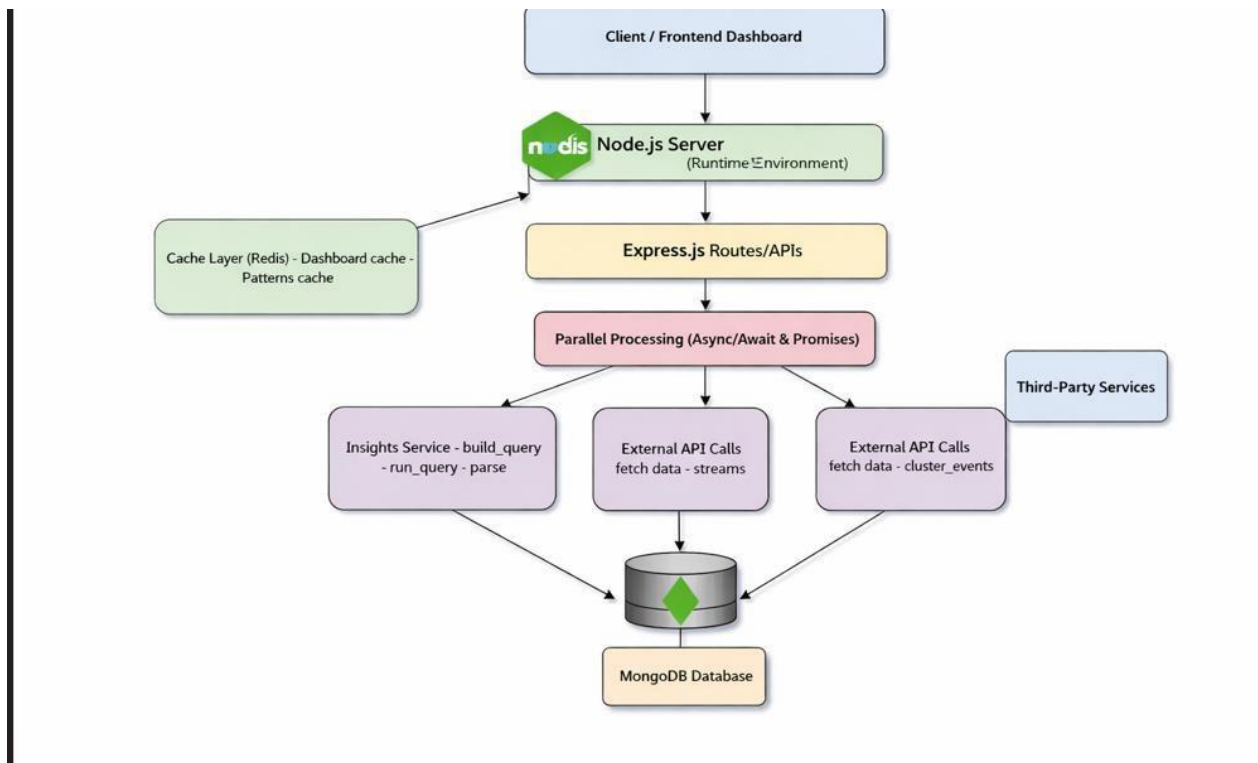
## **CHAPTER 3 INTERNSHIP DOMAIN**

### **3.1. INTRODUCTION**

The internship domain of Backend Development focuses on building and managing the server-side logic of web applications. It involves handling databases, creating APIs, managing authentication, and ensuring smooth communication between the frontend and backend systems.

During the internship, I worked as a Backend Developer, where I was responsible for developing and maintaining server-side functionalities for web applications. This included creating REST APIs, handling data flow between the client and server, and managing database operations.

The role also involved ensuring system performance, security, and scalability. Working in this domain helped me understand how real-world applications function behind the scenes and how different components of a system are connected to deliver a complete product.

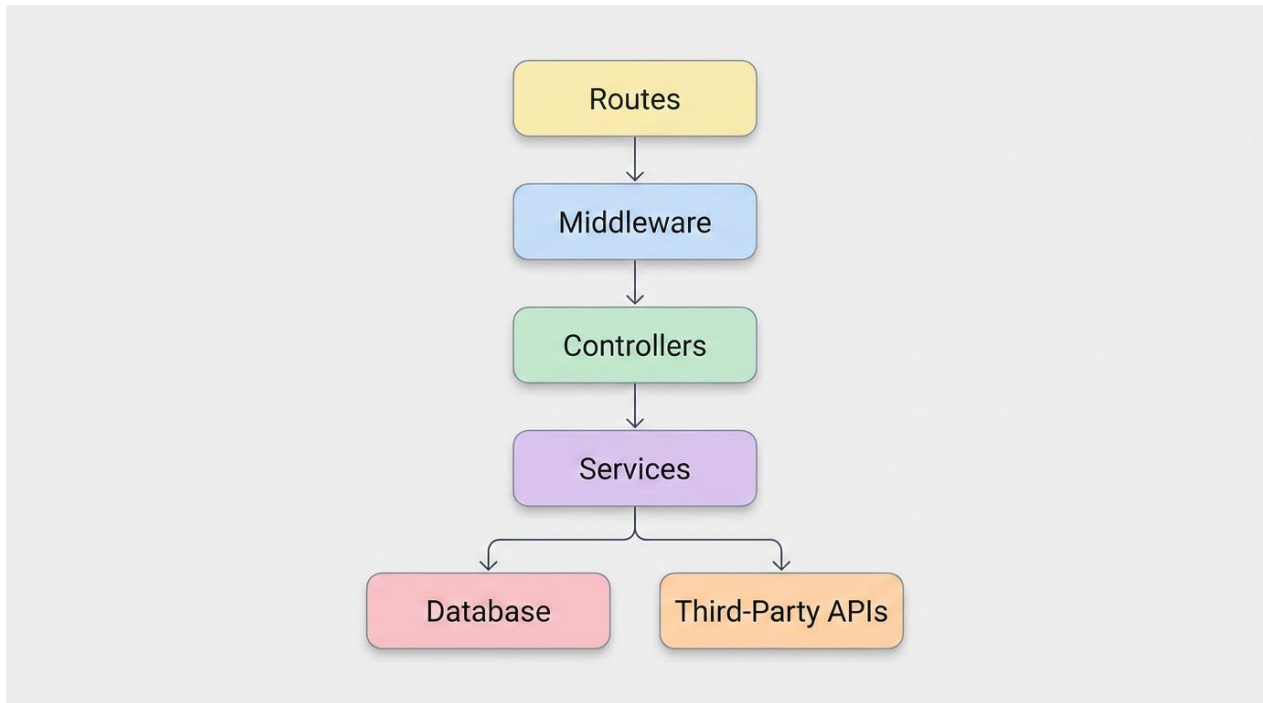


**Figure 3.1: Overall architecture**

## 3.2. TECHNOLOGIES USED

### 3.1.1 NodeJS

NodeJS was used as the runtime environment for executing JavaScript on the server side. It enabled efficient handling of server-side operations such as processing requests, managing asynchronous tasks, and building scalable backend



**Figure 3.2: Backend architecture for API handling**

### **3.2.1 React**

React was used to develop the frontend interface of the application. It provided a dynamic and responsive user experience for browsing and purchasing courses offered by the company

The frontend dashboard was designed to display key features such as available courses, course details, pricing information, user enrollments, and purchase history. React's component-based architecture helped in building reusable UI components like course cards, navigation bars, and dashboards, ensuring a clean, scalable, and efficient code structure for the application.

### **3.2.2 ExpressJs**

Express.js is commonly used to build routes, middleware, and APIs for web apps, and it is flexible enough to structure backend layers like routes, controllers, services, and database access.

In a course platform, Express.js is useful for connecting the frontend to the database, checking user access, handling payments, managing course content, and working with third-party services like email or file storage. It is a good choice because it is lightweight, fast, and flexible, so you can build a clean backend structure for features like authentication, course enrollment, admin panels, and protected video access.

### **3.2.4 MongoDB**

MongoDB was used as the database for the course selling website to store and manage application data in a flexible and scalable way. It is a NoSQL database that stores data in JSON-like documents, which makes it well-suited for handling dynamic data such as users, courses, enrollments, and transactions.

MongoDB was used to store user details, course information, payments, and purchase history. Its flexible schema made it easy to scale and update, while also ensuring fast and efficient data retrieval.

## **3.3 TOOLS USED**

### **3.3.1 Git And GitHub**

Git was used for version control to track changes in the project code and manage different versions efficiently. It helped in maintaining a history of code updates and allowed easy rollback when needed.

GitHub was used as a cloud-based platform to store and manage the project repository. It enabled collaboration, code sharing, and backup of the project, making it easier to work and maintain the course selling website.

### **3.3.2 Development Environment**

Modern development tools such as code editors and terminal interfaces were used for writing, testing, and debugging the application. These tools provided features like syntax highlighting, version control integration, and efficient debugging capabilities, which enhanced productivity during development.

### **3.3.3 Web Browser**

Web browsers were used to test the frontend application and visualize the dashboard outputs. They enabled real-time interaction with the system and helped in verifying the correctness of displayed data and user interface behavior.

### **3.3.4 API Testing Tools**

API testing tools were used to validate backend endpoints and ensure correct data flow between the frontend and backend systems. These tools helped in testing various scenarios, verifying responses, and debugging issues during development.

## **3.4 TESTING**

### **3.4.1 Unit Testing**

Unit testing was performed to verify the functionality of individual components such as API endpoints, data processing functions, and utility modules. Each component was tested separately to ensure correctness and reliability.

This helped in identifying errors at an early stage and ensured that every part of the system worked as expected before integration.

### **3.4.2 System Testing**

System testing was performed to validate the complete workflow of the application, from course browsing and user actions to data processing and display. It ensured that all components of the system worked together seamlessly.

This included testing the integration between the frontend, backend services, and database to verify the overall functionality and performance of the course selling website.

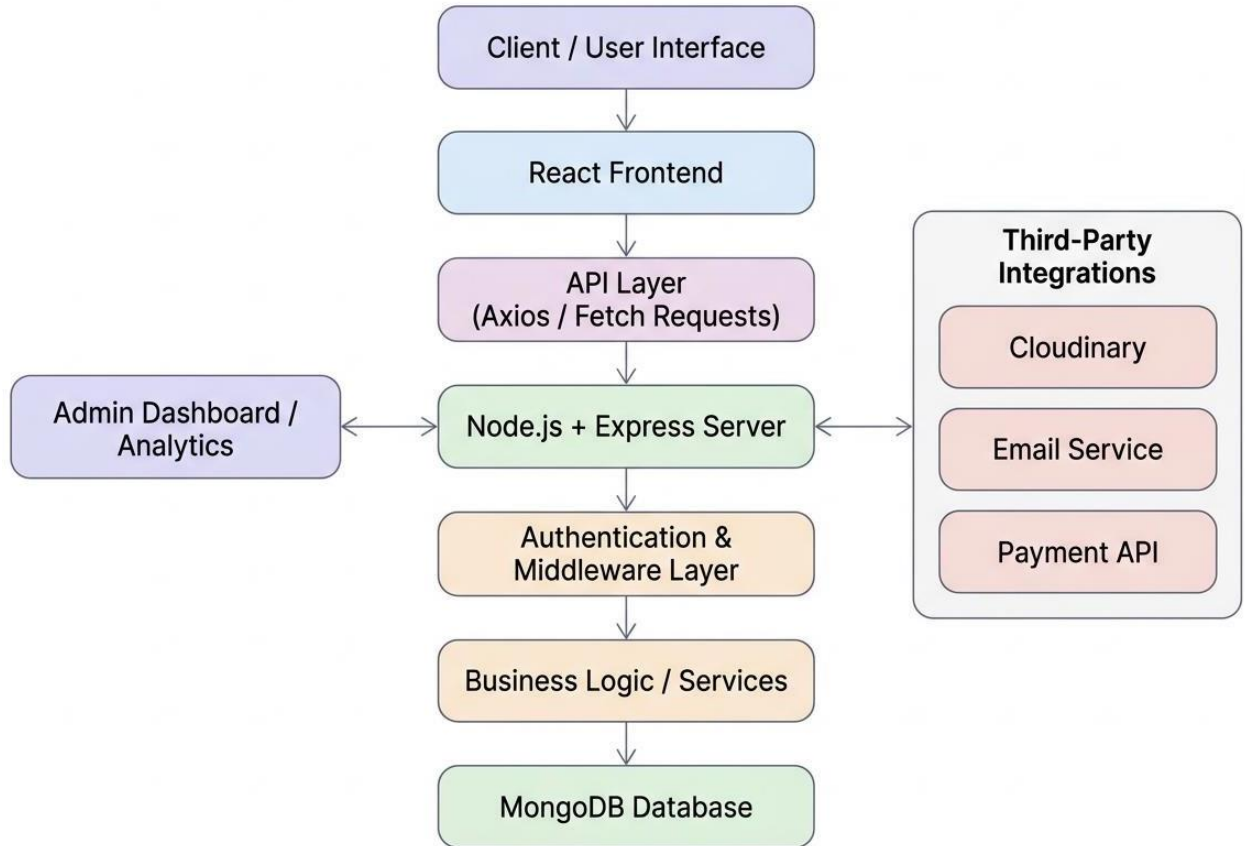
### **3.4.3 Performance Testing**

Performance testing was conducted to evaluate the speed, responsiveness, and stability of the course selling website under different workloads. It ensured that the application performs efficiently when multiple users access the system simultaneously.

This testing helped in identifying response time delays, optimizing API performance, and ensuring smooth user experience during high traffic conditions.

## CHAPTER 4 TASK PERFORMED

### 4.1. ABOUT THE PROJECT



**Figure 4.1: Overall system architecture of the website**

The project undertaken during the internship focused on the design and development of a scalable backend system for a course-selling web application. The primary objective was to build a robust platform capable of handling user interactions such as course browsing, enrolment, and secure transactions efficiently.

The system was developed using a backend powered by Node.js and Express.js, which handled API development, authentication, and business logic. A frontend interface built using React was used to provide an interactive and user-friendly experience for users. The application utilized MongoDB for efficient data storage and management of users, courses, and transactions.

The project emphasized efficient data handling and secure user authentication, particularly focusing on managing course data and user activity. This was achieved by designing RESTful APIs, implementing authentication mechanisms, and optimizing database queries. The system was designed with scalability and performance in mind, incorporating techniques such as modular architecture, efficient API handling, and optimized data flow.

## **4.2. PROOF OF CONCEPT**

Proof of Concept (PoC) was initially developed to validate the feasibility of building a backend system for a course-selling web application. This phase focused on establishing API communication, database connectivity, and verifying the structure of the data being processed and stored.

The PoC demonstrated the ability to:

- Register and authenticate users securely
- Create and manage course data dynamically
- Handle API requests and responses efficiently
- Store and retrieve data from the database in a structured format.

Based on the PoC results, it was confirmed that the system could effectively support core functionalities such as user management, course handling, and data processing. This laid the foundation for building a fully functional and scalable course-selling platform..



## **4.3. TASK ACCOMPLISHED**

During the internship, several key functionalities were successfully implemented as part of the project:

### **4.3.1. User Management and Authentication**

- Implemented user profile features such as profile image upload and profile completeness tracking.
- Developed secure authentication using OTP verification with SMTP mail services.
- Added input validations including mobile number verification to ensure data accuracy

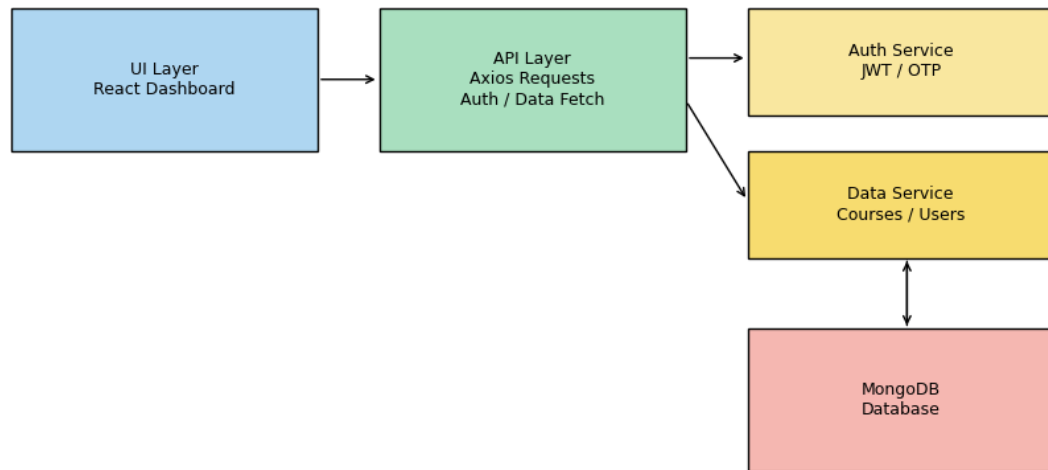
### **4.3.2 Admin Panel and Data Management**

- Designed logic to process raw log data and convert it into structured formats
- Implemented mechanisms to count log events corresponding to system operations, including medical record processing
- Applied filtering techniques to extract relevant logs based on keywords and conditions

### **4.3.3 System Optimization**

- Fixed UI issues such as mobile responsiveness, navbar bugs, and extra scrolling.
- Improved overall application performance by optimizing existing features

### 4.3.2. Dashboard Development

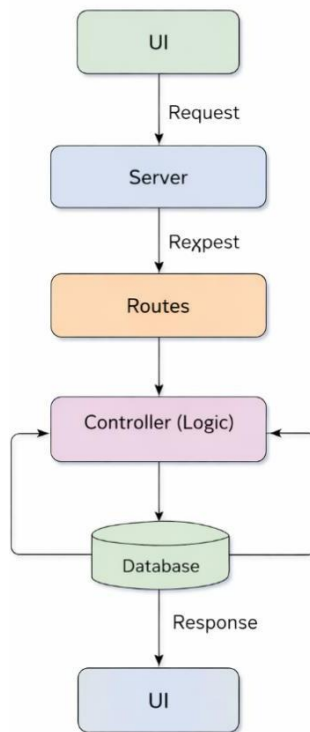


**Figure 4.2: Layered API Request–Response Flow for Dashboard System**

#### 4.3.4.1 Performance Optimization

- Optimized handling of large datasets such as attendance using Excel upload .
- Reduced unnecessary frontend operations to improve performance
- Enhanced UI responsiveness and smooth interaction.

#### 4.3.2.2. API Integration



**Figure 4.3: Api Integration**

- Developed RESTful APIs to connect backend services with the frontend interface
- Ensured smooth data flow and proper transformation of backend responses for frontend consumption
- Overall, the tasks helped in building a strong and efficient course selling system that can handle real-world user interactions smoothly.

## **4.4. INTERNSHIP WEEKLY REPORT**

Throughout the internship, progress was tracked on a weekly basis, documenting the tasks completed, challenges faced, and implemented solutions.

The weekly activities included:

- Understanding what the project needs to achieve and how the overall system is designed (its structure, components, and how they interact).
- Developing backend APIs for Data retrieval and Validations
- Implementing frontend components for User Interaction
- Testing and debugging Website functionalities
- Optimizing performance and improving Website scalability
- Collaborating with team members and receiving feedback from mentors

Regular communication with the team lead and project manager helped in aligning the work with project goals and ensuring timely completion of tasks



## 4.5. INTERNSHIP SUMMARY

As a backend developer, the internship provided hands-on experience in building cloud-based applications and handling large-scale data systems. It involved working on real-world backend challenges like processing high-volume data, improving API performance, and ensuring system reliability.

During the project, I gained practical knowledge in backend development, API design, cloud integration, and data handling. It also helped me improve my problem-solving skills and learn how to build efficient and scalable backend systems.

The successful development of the log analytics system, including tracking and analyzing medical record processing through logs, reflects the practical impact of my backend development work during the internship.

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# CHAPTER 5 INTERNSHIP OUTCOMES

## 5.1. INTERNSHIP OUTCOMES

The internship provided a comprehensive learning experience by combining theoretical knowledge with practical implementation in a real-world backend development project for a course selling website. The outcomes of the internship can be categorized into technical, analytical, and professional skill development, including API development, database handling, system design, and improved problem-solving abilities.

### 5.1.1. Technical Outcomes

During the internship, strong proficiency was developed in backend development using the **MERN stack (MongoDB, Express.js, React.js, Node.js)**. The project provided hands-on experience in building a course selling website with scalable backend architecture, API development, and database management. It also enhanced understanding of how frontend and backend interact in a full-stack application.

Key technical achievements include:

- Designing and developing RESTful APIs for user, course, and order management
- Implementing secure authentication and authorization for users
- Managing and integrating MongoDB database for efficient data storage
- Developing backend logic for course creation, purchase, and access control
- Optimizing server-side performance for smooth application flow
- Integrating frontend with backend for dynamic data handling

### 5.1.2. Analytical Outcomes

The internship significantly enhanced analytical and problem-solving skills through the design and development of a **course selling website using the MERN stack**. The ability to understand user behavior, system workflows, and data flow between frontend and backend was a key learning outcome.

Key analytical contributions include:

- Identifying user interaction patterns across the course selling platform
- Analyzing course purchase flow and optimizing user experience
- Classifying and managing data such as users, courses, and orders based on system requirements
- Evaluating API performance and improving response efficiency
- Analyzing backend logs and request flow to ensure system reliability and smooth functionality

### 5.1.3. Professional Outcomes

In addition to technical skills, the internship helped in developing essential professional competencies required in a corporate environment.

These include:

- Effective communication with team members and mentors
- Understanding project workflows and development life cycles
- Time management and task prioritization
- Adapting to real-world problem-solving scenarios
- Gaining exposure to collaborative software development practices



## 5.2. CONCLUSION

The internship was a valuable and enriching experience that provided deep insights into the development of a **course selling website using the MERN stack**. The project demonstrated how a complete web application can be built by integrating frontend and backend technologies to manage users, courses, and transactions efficiently.

By leveraging technologies such as **MongoDB, Express.js, React.js, and Node.js**, a scalable and functional course selling platform was developed. The system effectively supports user authentication, course management, and seamless data flow between client and server.

A key highlight of the project was the implementation of backend logic for handling course creation, purchase workflows, and secure access control, showcasing the practical application of full-stack development in real-world scenarios. The optimization of APIs and database operations further improved system performance and reliability.

Overall, the internship bridged the gap between academic learning and industry practices, providing hands-on experience in building scalable web applications. The knowledge and skills gained during this period will serve as a strong foundation for future growth in full-stack and backend development.

# **BIBLIOGRAPHY**

1. React Documentation – Official Website
2. Node.js Documentation – Official Website
3. Express.js Documentation – Official Website
4. MongoDB Documentation – Official Website
5. Axios Documentation – Official Website
6. Docker Documentation – Official Website

# ENCLOSURES

## Enclosure 1: Offer Letter

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>REBERT TECHNOLOGIES PVT. LTD.</b><br>Address: Yerwada, Pune, Maharashtra, India 411006<br>CIN: U72900PN202IPTC20793   rebertnextgen.com   contact@rebertnextgen.com |  |
| <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                        |                                                                                     |
| <b>OFFER LETTER</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                        |                                                                                     |
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| <p>To,<br/>Mr. Ravi Tharun<br/>Email: tharunravi672@gmail.com</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                        |                                                                                     |
| <p><b>Subject: Offer Letter for Backend Developer</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                        |                                                                                     |
| <p><b>Dear Ravi Tharun,</b></p> <p>On behalf of <b>Rebert Technologies Pvt. Ltd.</b> We are delighted to offer you the position of <b>Backend Developer</b> with Rebert Technologies Pvt. Ltd. We believe that you will contribute significantly to the success of our organization, and we look forward to a long and mutually rewarding association.<br/>Your appointment will be governed by the terms and conditions outlined in the annexures attached to this letter:</p> <ul style="list-style-type: none"><li>• <b>Annexure A</b> – Compensation &amp; Benefits</li><li>• <b>Annexure B</b> – Company Policies &amp; Code of Conduct</li><li>• <b>Annexure C</b> – Confidentiality &amp; Non-Compete Agreement</li></ul> <p>Please review the details carefully. To confirm your acceptance of this offer, kindly sign and return a copy of this letter along with the annexures by <b>26-Nov-2025</b>.</p> <p>We are excited to welcome you to our team and wish you every success in your career with <b>Rebert Technologies Pvt. Ltd. Congratulations!</b></p> <p>Please sign and return a copy of this letter as a token of your acceptance.</p> <div style="text-align: center;"><br/></div> <p>Ms. Sara Shaikh<br/>Chief Technology Officer,<br/>Rebert Technologies Pvt. Ltd.<br/>Pune, Maharashtra, 411006<br/>Date: 26-Nov-2025</p> |                                                                                                                                                                        |                                                                                     |
| <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                        |                                                                                     |
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# ENCLOSURES

## Enclosure 2: Internship Certificate:

